CHAPTER 2
WORKING WITH FINANCIAL STATEMENTS

Answers to Concepts Review and Critical Thinking Questions

1. Liquidity measures how quickly and easily an asset can be converted to cash without significant loss in value. It's desirable for firms to have high liquidity so that they can more safely meet short-term creditor demands. However, liquidity also has an opportunity cost. Firms generally reap higher returns by investing in illiquid, productive assets. It's up to the firm's financial management staff to find a reasonable compromise between these opposing needs.

2. The recognition and matching principles in financial accounting call for revenues, and the costs associated with producing those revenues, to be "booked" when the revenue process is essentially complete, not necessarily when the cash is collected or bills are paid. Note that this way is not necessarily correct; it's the way accountants have chosen to do it.

3. Historical costs can be objectively and precisely measured, whereas market values can be difficult to estimate, and different analysts would come up with different numbers. Thus, there is a tradeoff between relevance (market values) and objectivity (book values).

4. Depreciation is a non-cash deduction that reflects adjustments made in asset book values in accordance with the matching principle in financial accounting. Interest expense is a cash outlay, but it's a financing cost, not an operating cost.

5. Market values can never be negative. Imagine a share of stock selling for –$20. This would mean that if you placed an order for 100 shares, you would get the stock along with a check for $2,000. How many shares do you want to buy? More generally, because of corporate and individual bankruptcy laws, net worth for a person or a corporation cannot be negative, implying that liabilities cannot exceed assets in market value.

6. For a successful company that is rapidly expanding, capital outlays would typically be large, possibly leading to negative cash flow from assets. In general, what matters is whether the money is spent wisely, not whether cash flow from assets is positive or negative.

7. It's probably not a good sign for an established company, but it would be fairly ordinary for a start-up, so it depends.

8. For example, if a company were to become more efficient in inventory management, the amount of inventory needed would decline. The same might be true if it becomes better at collecting its receivables. In general, anything that leads to a decline in ending NWC relative to beginning NWC would have this effect. Negative net capital spending would mean more long-lived assets were liquidated than purchased.

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9. If a company raises more money from selling stock than it pays in dividends in a particular period, its cash flow to stockholders will be negative. If a company borrows more than it pays in interest, its cash flow to creditors will be negative.

10. The adjustments discussed were purely accounting changes; they had no cash flow or market value consequences unless the new accounting information caused stockholders to revalue the company.

Solutions to Questions and Problems

NOTE: All end-of-chapter problems were solved using a spreadsheet. Many problems require multiple steps. Due to space and readability constraints, when these intermediate steps are included in this solutions manual, rounding may appear to have occurred. However, the final answer for each problem is found without rounding during any step in the problem.

Basic

1. To find owners' equity, we must construct a balance sheet as follows:

Balance Sheet			
CA	$ 5,200	CL	$ 4,100
NFA	27,300	LTD	10,900
		OE	??
TA	$32,500	TL & OE	$32,500

We know that total liabilities and owners' equity (TL & OE) must equal total assets of $32,500. We also know that TL & OE is equal to current liabilities plus long-term debt plus owners' equity, so owners' equity is:

Owners' equity = $32,500 – 10,900 – 4,100
Owners' equity = $17,500

And net working capital (NWC) is:

NWC = CA – CL
NWC = $5,200 – 4,100
NWC = $1,100

2. The income statement for the company is:

Income Statement	
Sales	$757,000
Costs	316,000
Depreciation	41,000
EBIT	$400,000
Interest	36,000
EBT	$364,000
Taxes (21%)	76,440
Net income	$287,560

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3. One equation for net income is:

Net income = Dividends + Addition to retained earnings

Rearranging, we get:

Addition to retained earnings = Net income – Dividends = $287,560 – 125,000 = $162,560

4. EPS = Net income/Shares = $287,560/75,000 = $3.83 per share

DPS = Dividends/Shares = $125,000/75,000 = $1.67 per share

5. Taxes = .10($11,600) + .12($47,150 – 11,600) + .22($100,525 – 47,150) + .24($191,950 – 100,525)
+ .32($195,000 – 191,950)
Taxes = $40,086.50

The average tax rate is the total tax paid divided by taxable income, so:

Average tax rate = $40,086.50/$195,000
Average tax rate = .2056, or 20.56%

The marginal tax rate is the tax rate on the next dollar of income. The person has net income of $195,000 and the 32 percent tax bracket is applicable to a taxable income up to $243,725, so the marginal tax rate is 32 percent.

6. To calculate OCF, we first need the income statement:

Income Statement	
Sales	$63,300
Costs	24,200
Depreciation	2,300
EBIT	$36,800
Interest	1,800
Taxable income	$35,000
Taxes (22%)	7,700
Net income	$27,300

OCF = EBIT + Depreciation – Taxes
OCF = $36,800 + 2,300 – 7,700
OCF = $31,400

7. Net capital spending = $NFA_{end} – NFA_{beg}$ + Depreciation

Net capital spending = $3,200,000 – 2,450,000 + 347,000
Net capital spending = $1,097,000

8. Change in NWC = $NWC_{end} – NWC_{beg}$
Change in NWC = $(CA_{end} – CL_{end}) – (CA_{beg} – CL_{beg})$
Change in NWC = ($6,030 – 3,250) – ($5,530 – 2,610)
Change in NWC = $2,780 – 2,920
Change in NWC = –$140

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9. Cash flow to creditors = Interest paid – Net new borrowing
Cash flow to creditors = Interest paid – (LTD_{end} – LTD_{beg})
Cash flow to creditors = $265,000 – ($2,730,000 – 2,400,000)
Cash flow to creditors = –$65,000

10. Cash flow to stockholders = Dividends paid – Net new equity
Cash flow to stockholders = Dividends paid – [($Common_{end}$ + $APIS_{end}$) – ($Common_{beg}$ + $APIS_{beg}$)]
Cash flow to stockholders = $654,000 – [($975,000 + 5,030,000) – ($770,000 + 4,890,000)]
Cash flow to stockholders = $309,000

Note, APIS is the additional paid-in surplus.

11. Cash flow from assets = Cash flow to creditors + Cash flow to stockholders
= –$65,000 + 309,000 = $244,000

Cash flow from assets = $244,000 = OCF – Change in NWC – Net capital spending
= $244,000 = OCF – (–$55,000) – 1,500,000

Operating cash flow = $244,000 – 55,000 + 1,500,000
Operating cash flow = $1,689,000

Intermediate

12. To find the book value of current assets, we use: NWC = CA – CL. Rearranging to solve for current assets, we get:

CA = NWC + CL
CA = $375,000 + 945,000
CA = $1,320,000

The market value of current assets and fixed assets is given, so:

Book value CA	= $1,320,000	Market value NWC	= $1,250,000
Book value NFA	= 3,700,000	Market value NFA	= 5,800,000
Book value assets	= $5,020,000	Total	= $7,050,000

13. To find the OCF, we first calculate net income.

Income Statement	
Sales	$373,000
Costs	200,300
Other expenses	10,200
Depreciation	21,900
EBIT	$140,600
Interest	15,700
Taxable income	$124,900
Taxes	26,299
Net income	$ 98,601
Dividends	$21,450
Additions to RE	$77,151

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a. OCF = EBIT + Depreciation – Taxes
OCF = $140,600 + 21,900 – 26,299
OCF = $136,201

b. CFC = Interest – Net new LTD
CFC = $15,700 – (–5,400)
CFC = $21,100

Note that the net new long-term debt is negative because the company repaid part of its long-term debt.

c. CFS = Dividends – Net new equity
CFS = $21,450 – 7,300
CFS = $14,150

d. We know that CFA = CFC + CFS, so:

CFA = $21,100 + 14,150
CFA = $35,250

CFA is also equal to OCF – Net capital spending – Change in NWC. We already know OCF. Net capital spending is equal to:

Net capital spending = Increase in NFA + Depreciation
Net capital spending = $75,600 + 21,900
Net capital spending = $97,500

Now we can use:

CFA = OCF – Net capital spending – Change in NWC
$35,250 = $136,201 – 97,500 – Change in NWC
Change in NWC = $3,451

This means that the company increased its NWC by $3,451.

14. The solution to this question works the income statement backwards. Starting at the bottom:

Net income = Dividends + Addition to retained earnings = $2,370 + 6,900 = $9,270

Now, looking at the income statement:

EBT – EBT × Tax rate = Net income

Recognize that EBT × Tax rate is the calculation for taxes. Solving this for EBT yields:

EBT = NI/(1 – Tax rate) = $9,270/(1 – .22) = $11,885

Now you can calculate:

EBIT = EBT + Interest = $11,885 + 5,800 = $17,685

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The last step is to use:

EBIT = Sales – Costs – Depreciation
$17,685 = $79,400 – 53,900 – Depreciation
Depreciation = $7,815

15. The balance sheet for the company looks like this:

Balance Sheet			
Cash	$ 155,000	Accounts payable	$ 273,000
Accounts receivable	157,000	Notes payable	203,800
Inventory	384,000	Current liabilities	$ 476,800
Current assets	$ 696,000	Long-term debt	1,079,000
		Total liabilities	$1,555,800
Tangible net fixed assets	$2,130,000		
Intangible net fixed assets	903,000	Common stock	??
		Accumulated ret. earnings	1,778,000
Total assets	$3,729,000	Total liab. & owners' equity	$3,729,000

Total liabilities and owners' equity is:

TL & OE = CL + LTD + Common stock + Retained earnings

Solving this equation for common stock gives us:

Common stock = $3,729,000 – 1,778,000 – 1,555,800
Common stock = $395,200

16. The market value of shareholders' equity cannot be negative. A negative market value in this case would imply that the company would pay you to own the stock. The market value of shareholders' equity can be stated as: Shareholders' equity = Max[(TA – TL), 0]. So, if TA are $12,900, equity is equal to $1,300, and if TA are $9,400, equity is equal to $0. We should note here that the book value of shareholders' equity can be negative.

17.

Income Statement	
Sales	$835,000
COGS	525,000
A&S expenses	120,000
Depreciation	165,000
EBIT	$25,000
Interest	90,000
Taxable income	–$65,000
Taxes (25%)	0
a. Net income	–$65,000

b. OCF = EBIT + Depreciation – Taxes
OCF = $25,000 + 165,000 – 0
OCF = $190,000

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c. Net income was negative because of the tax deductibility of depreciation and interest expense. However, the actual cash flow from operations was positive because depreciation is a non-cash expense and interest is a financing expense, not an operating expense.

18. A firm can still pay out dividends if net income is negative; it just has to be sure there is sufficient cash flow to make the dividend payments.

Change in NWC = Net capital spending = Net new equity = 0 (Given)
Cash flow from assets = OCF – Change in NWC – Net capital spending
Cash flow from assets = $190,000 – 0 – 0 = $190,000
Cash flow to stockholders = Dividends – Net new equity = $128,000 – 0 = $128,000

Cash flow to creditors = Cash flow from assets – Cash flow to stockholders
Cash flow to creditors = $190,000 – 128,000 = $62,000
Cash flow to creditors = Interest – Net new LTD
Net new LTD = Interest – Cash flow to creditors = $90,000 – 62,000 = $28,000

19. *a.*

Income Statement	
Sales	$45,686
Cost of goods sold	32,602
Depreciation	8,111
EBIT	$ 4,973
Interest	3,660
Taxable income	$ 1,313
Taxes (22%)	289
Net income	$ 1,024

b. OCF = EBIT + Depreciation – Taxes
= $4,973 + 8,111 – 289 = $12,795

c. Change in NWC $= NWC_{end} - NWC_{beg}$
$= (CA_{end} - CL_{end}) - (CA_{beg} - CL_{beg})$
= ($11,885 – 6,313) – ($9,630 – 5,413)
= $1,355

Net capital spending $= NFA_{end} - NFA_{beg} + Depreciation$
= $33,664 – 27,348 + 8,111
= $14,427

CFA = OCF – Change in NWC – Net capital spending
= $12,795 – 1,355 – 14,427
= –$2,987

The cash flow from assets can be positive or negative, since it represents whether the firm raised funds or distributed funds on a net basis. In this problem, even though net income and OCF are positive, the firm invested heavily in both fixed assets and net working capital; it had to raise a net $2,987 in funds from its stockholders and creditors to make these investments.

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d. Cash flow to creditors = Interest – Net new LTD = \$3,660 – 0 = \$3,660

Cash flow to stockholders = Cash flow from assets – Cash flow to creditors
= –\$2,987 – 3,660 = –\$6,647

We can also calculate the cash flow to stockholders as:

Cash flow to stockholders = Dividends – Net new equity

Solving for net new equity, we get:

Net new equity = \$2,604 – (–6,647) = \$9,251

The firm had positive earnings in an accounting sense (NI > 0) and had positive cash flow from operations. The firm invested \$1,355 in new net working capital and \$14,427 in new fixed assets. The firm had to raise \$2,987 from its stakeholders to support this new investment. It accomplished this by raising \$9,251 in the form of new equity. After paying out \$2,604 of this in the form of dividends to shareholders and \$3,660 in the form of interest to creditors, \$2,987 was left to meet the firm's cash flow needs for investment.

20. *a.* Total assets_{2024} = \$1,005 + 4,144 = \$5,149

$\text{Total liabilities}_{2024}$ = \$402 + 2,190 = \$2,592
$\text{Owners' equity}_{2024}$ = \$5,149 – 2,592 = \$2,557

$\text{Total assets}_{2025}$ = \$1,089 + 4,990 = \$6,079
$\text{Total liabilities}_{2025}$ = \$451 + 2,329 = \$2,780
$\text{Owners' equity}_{2025}$ = \$6,079 – 2,780 = \$3,299

b. $NWC_{2024} = CA_{2024} – CL_{2024}$ = \$1,005 – 402 = \$603
$NWC_{2025} = CA_{2025} – CL_{2025}$ = \$1,089 – 451 = \$638
Change in NWC = $NWC_{2025} – NWC_{2024}$ = \$638 – 603 = \$35

c. We can calculate net capital spending as:
Net capital spending = $\text{Net fixed assets}_{2025} – \text{Net fixed assets}_{2024}$ + Depreciation
Net capital spending = \$4,990 – 4,144 + 1,136 = \$1,982

So, the company had a net capital spending cash flow of \$1,982. We also know that net capital spending is:

Net capital spending = Fixed assets bought – Fixed assets sold
\$1,982 = \$2,080 – Fixed assets sold
Fixed assets sold = \$2,080 – 1,982 = \$98

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To calculate the cash flow from assets, we must first calculate the operating cash flow. The income statement is:

Income Statement	
Sales	$12,751
Costs	5,946
Depreciation expense	1,136
EBIT	$ 5,669
Interest expense	323
EBT	$ 5,346
Taxes (21%)	1,123
Net income	$ 4,223

So, the operating cash flow is:

OCF = EBIT + Depreciation – Taxes = $5,669 + 1,136 – 1,123 = $5,682

And the cash flow from assets is:

Cash flow from assets = OCF – Change in NWC – Net capital spending
= $5,682 – 35 – 1,982 = $3,665

d. Net new borrowing = $LTD_{2025} – LTD_{2024}$ = $2,329 – 2,190 = $139
Cash flow to creditors = Interest – Net new LTD = $323 – 139 = $184

Net new borrowing = $139 = Debt issued – Debt retired
Debt retired = $420 – 139 = $281

21. To construct the cash flow identity, we will begin with cash flow from assets. Cash flow from assets is:

Cash flow from assets = OCF – Change in NWC – Net capital spending

So, the operating cash flow is:

OCF = EBIT + Depreciation – Taxes
OCF = $81,741 + 72,489 – 14,028
OCF = $140,202

Next, we will calculate the change in net working capital, which is:

Change in NWC = $NWC_{end} – NWC_{beg}$
Change in NWC = $(CA_{end} – CL_{end}) – (CA_{beg} – CL_{beg})$
Change in NWC = ($77,250 – 35,834) – ($61,240 – 31,870)
Change in NWC = $12,046

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Now, we can calculate the net capital spending. The capital spending is:

Net capital spending = $NFA_{end} – NFA_{beg}$ + Depreciation
Net capital spending = $539,679 – 457,454 + 72,489
Net capital spending = $154,714

Now, we have the cash flow from assets, which is:

Cash flow from assets = OCF – Change in NWC – Net capital spending
Cash flow from assets = $140,202 – 12,046 – 154,714
Cash flow from assets = –$26,558

The company's assets generated an outflow of $26,558. The cash flow from operations was $140,202, and the company spent $12,046 on net working capital and $154,714 on fixed assets.

The cash flow to creditors is:

Cash flow to creditors = Interest paid – New long-term debt
Cash flow to creditors = Interest paid – (Long-term $debt_{end}$ – Long-term $debt_{beg}$)
Cash flow to creditors = $25,630 – ($201,900 – 181,000)
Cash flow to creditors = $4,730

The cash flow to stockholders is a little trickier in this problem. First, we need to calculate the new equity sold. The equity balance increased during the year. The only way to increase the equity balance is through retained earnings or selling equity. To calculate the new equity sold, we can use the following equation:

New equity = Ending equity – Beginning equity – Addition to retained earnings
New equity = $379,195 – 305,824 – 27,883
New equity = $45,488

What happened was the equity account increased by $73,371. Of this increase, $27,883 came from addition to retained earnings, so the remainder must have been the sale of new equity. Now we can calculate the cash flow to stockholders as:

Cash flow to stockholders = Dividends paid – Net new equity
Cash flow to stockholders = $14,200 – 45,488
Cash flow to stockholders = –$31,288

The company paid $4,730 to creditors and raised $31,288 from stockholders.

Finally, the cash flow identity is:

Cash flow from assets = Cash flow to creditors + Cash flow to stockholders
–$26,558 = $4,730 + –$31,288

The cash flow identity balances, which is what we expect.

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Challenge

22. Net capital spending $= NFA_{end} - NFA_{beg} + Depreciation$

$= (NFA_{end} - NFA_{beg}) + (Depreciation + AD_{beg}) - AD_{beg}$
$= (NFA_{end} - NFA_{beg}) + AD_{end} - AD_{beg}$
$= (NFA_{end} + AD_{end}) - (NFA_{beg} + AD_{beg})$
$= FA_{end} - FA_{beg}$

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